

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 11)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
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7	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain how climate change may affect the survival of species. Use named examples.

[6 marks]

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0 1 . 2

Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

[2 marks]

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0 1 . 3

Explain why polar regions warming faster than the equator affects the weather in the UK.

[2 marks]

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10

0 2 . 1 Explain how each of the following human activities increases soil erosion.
[4 marks]

Overgrazing

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Reducing soil biota

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0 2 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.
[2 marks]

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0 2 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.
[4 marks]

1

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2

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3

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4

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10

03 . 1

Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station

☐

B Tropospheric ozone

☐

C Carbon monoxide from a car exhaust

☐

D Nitrogen monoxide from a diesel engine

☐

03 . 2

Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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04 . 1

Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

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0 4 . 2

Compare **three** methods of storing hydrogen, giving a disadvantage of each.**[3 marks]**

0 4 . 3

Explain how a fuel cell differs from a rechargeable battery.

[2 marks]

0 4 . 4

Explain what is meant by the 'hydrogen economy'.

[1 mark]**10**

0 5 . 1

Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

0 5 . 2 Give **three** factors that affect the choice of disposal method for a solid waste. **[3 marks]**

1

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2

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3

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0 5 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill. **[2 marks]**

Advantage

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Disadvantage

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0 5 . 4 Explain why mixed municipal waste is difficult to recycle. **[1 mark]**

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10

0 6

Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

0 6**. 1**

Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]**Answer**

%**0 6****. 2**

Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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0 6**. 3**

Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

1

2

3

0 6 . 4 Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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10

0 7 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

07 . 1

Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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07 . 2

Explain the purpose of aeration (stage 3).

[2 marks]

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07 . 3

Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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07 . 4

Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic)

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Salt

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Odours and pesticides

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10

08 . 1

Explain why organic pollution of rivers is usually monitored by measuring Biochemical Oxygen Demand (BOD) rather than the concentrations of the pollutants themselves.

[3 marks]

08 . 2

Explain why a coliform count is used to assess the health risk from sewage contamination rather than testing for pathogens such as typhoid.

[2 marks]

08 . 3

Give **three** characteristics of species that make them useful in a biotic index.

[3 marks]

1

2

3

08 . 4

Give **one** advantage and **one** disadvantage of using a biotic index rather than chemical monitoring.

[2 marks]

Advantage

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Disadvantage

.....

10

0 9

Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

0 9

1

Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

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0 9

2

Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer

%

09 . 3

Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

1

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2

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09 . 4

Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

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10

Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

ESP/7447/1/SET11

Extra space

10 . **2** State **one** effect of tropospheric ozone on plants.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

1 1 . 2

Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

END OF QUESTIONS